Seth Baird

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Work Experience

- **SMS InfoComm (Wistron Corporation)** Dallas, TX
  NVIDIA Server Repair Technician *May 2024 - August 2024*
  - **System Repair and Troubleshooting** Diagnosed, refurbished, repaired, and built NVIDIA HGX 8-GPU units.
  - **Hardware Validation and Analysis** Used proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGAs before executing a repair.
  - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of by-hand data entry into the company's antiquated database, saving my team 80 man-hours per week
- **Jawa.gg** Remote
  Verified Seller *October 2021 - August 2023*
  - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
  - **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
  - **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.

Involvement

- **Purdue Electric Racing - STM32 Software Team**
  - Created multiple HALs (Hardware abstraction layers) to simplify the codebase and make development easier
  - Reworked the CAN communication backend to use two busses simultaneously for new high-throughput motors
  - Utilized testing equipment such as CANable and the oscilloscope's built in signal decoding to verify new hardware

Skills and Projects

- **Hardware Engineering**
  - Implemented a channel scan algorithm into a hand-built superheterodyne FM radio using DSP and statistics
  - Fabricated a Li-ion powered guitar amplifier with a custom case and purpose-built PCBs laid out in KiCad
  - Leveraged Solidworks, OpenSCAD, and a Raspberry Pi Pico running Rust to create a replica arcade game controller
  - Added smart capability to a ceiling fan using an ESP8266 Wi-Fi microcontroller (Arduino C) and the Blynk API
  - Designed and simulated logic circuits and HDL code for Altera FPGAs in Quartus.
- **IT and Software Development**
  - Developed an android app that integrated with a car's engine computer to provide data and insights to the driver
  - Created over 20 proof-of-concept iOS and Android apps to gain familiarity with Swift and Kotlin
  - Deployed a bash script using cron to automate bidirectional encrypted data transfer for on- and off-site backups
  - Implemented a domain filtering DNS server to block advertisements and malicious sites on all devices

Education

- **Purdue University** West Lafayette, Indiana
  In progress - Anticipated Graduation May 2027

  Major in Computer Engineering Technology *Minor in Information Technology*
- **Memorial High School** Frisco, Texas
  Graduated May 2023

  Vocational Education: *CS3: Data Structures and Algorithms, Mobile Application Development*